Machine Learning with Applications 16 (2024) 100547

Contents lists available at ScienceDirect

Machine Learning with Applications

journal homepage: www.elsevier.com/locate/mlwa

Augmenting roadway safety with machine learning and deep learning: Pothole detection and dimension estimation using in-vehicle technologies

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ARTICLE INFO

Keywords:
YOLOv5
Vehicle built-in technologies
Pavement condition monitoring
Pothole size
Computer vision
Pavement maintenance

ABSTRACT

Detection and estimation of pothole dimensions is an essential step in road maintenance. Aging, heavy rainfall, traffic, and weak underlying layers may cause pavement potholes. Potholes can cause accidents when drivers lose control after hitting or swerving to avoid them, which may lead to injuries or fatal crashes. Also, potholes may result in property damages, such as flat tires, scrapes, dents, and leaks. Additionally, potholes are costly; for example, in the United States, potholes cost drivers about $3 Billion annually. Traditional ways of attending to potholes involve field surveys carried out by skilled personnel to determine their sizes for quantity and cost estimates. This process is expensive, prone to errors, subjectivity, unsafe, and time-consuming. Some authorities use sensor vehicles to carry out the surveys, a method that is accurate, safer, and faster than the traditional approach but much more expensive; therefore, not all authorities can afford them. To avoid these challenges, a modern, real-time, cost-effective approach is proposed to ensure the efficient and fast process of pothole maintenance. This paper presents a Deep Learning model trained using the You Only Look Once (YOLO) algorithm to capture potholes and estimate their dimensions and locations using only built-in vehicle technologies. The model attained 93.0 % precision, 91.6 % recall, 87.0 % F1-score, and 96.3 % mAP. A statistical analysis of the on-site test results indicates that the results are significant at a 5 % level, with a p-value of 0.037. This approach provides an economical and faster way of monitoring road surface conditions.

Introduction

Paved roads deteriorate over time due to various factors, including loading, weather changes (temperature changes), poor mix designs, poor construction techniques (inadequate compaction), high water tables, poor drainage, and poor quality of materials (Pavement Interactive, 2020). As a result, paved roads form various distresses such as fatigue/alligator cracks, block cracks, transverse cracks, longitudinal-wheel path cracks, longitudinal non–wheel path cracks, edge, joint, and reflective cracks, patches, potholes, raveling, shoving, and rutting (John S. Miller & William Y. Bellinger, 2014). Early detection of distress and maintenance are critical for improving roadways' safety and operational efficiency. Delaying the maintenance processes causes increases in deterioration, which in turn causes increased maintenance costs (Majidifard et al., 2020). Also, delayed maintenance could result in death or severe injury, contributing to increased traffic fatalities which are high (Chengula et al., 2023a). Therefore, it is necessary to monitor road conditions continuously for distress formation and maintenance (Fan et al., 2022) while ensuring safety for all road users (Ngeni et al., 2022).

This paper deals with the detection and size estimation of potholes only as part of road condition monitoring (RCM), an essential part of road maintenance and rehabilitation (M&R) (Sahin et al., 2014). In cases where the authorities encounter limitations (such as resources), the attendance to pavement distress depends on their classification. The order of priority follows the order: safety, structural, and functional (AASHTO, 2020; FHWA, 2012). Potholes classified under the safety category are given priority for maintenance. Therefore, this study focuses on the detection, size estimation, and geolocating of potholes.

Potholes are bowl-shaped holes of various sizes on the pavement surface, with a minimum plan dimension of 150 mm (John S. Miller & William Y. Bellinger, 2014). They pose a danger to various road users, and it is estimated that the United States government spent about $3 billion in 2017 to fix them (ACSC, 2023). In 2021 alone, the survey shows that potholes cost American drivers $26.5 billion, with an average

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https://doi.org/10.1016/j.mlwa.2024.100547
Received 1 December 2023; Received in revised form 9 March 2024; Accepted 12 March 2024
Available online 13 March 2024

cost of $600 per repair of car damage caused after hitting potholes (Edmonds, 2012). Concerning traffic safety, potholes were reported to cause about 3597 deaths in India in 2017 (T. E. Times, 2018). Potholes also affect pedestrians, and it is estimated that about 2015 pedestrians lost their lives in India in 2018 alone (Sabha, 2019).

Numerous methods exist in the literature for the detection and size estimation of potholes. These methods are grouped into two broad categories: manual and automated. Manual methods involve using survey forms, where surveyors or experienced experts move along roads, inspect, take measurements, and record for further consideration. The technique is labor-intensive, requires much time, and poses a danger to surveyors and other road users therefore unsafe (Ajmera et al., 2022; Fares & Zayed, 2023; Majidifard et al., 2020). This approach is dependent on individual experience. Therefore, it leads to inaccurate results (Asad et al., 2022). Some state Departments of Transportation in the United States have established online pothole-reporting systems to reduce the burden and expenses. Through these systems, the traveling public shares information regarding potholes' presence in various parts of roadways (Devine, 2017; Wong, 2022). Automated methods use special survey vehicles with 3D laser scanners and vibration-based technologies. The methods are not cost-effective because they require special vehicles (Motwani & Sharma, 2020). Using computer vision (CV) techniques is promising as the future of data collection for pavement maintenance and rehabilitation. A study by Ngeni et al. (2024) shows that quantum computing (QC) can be used to improve the performance of CV models when too much computing power is needed. Notwithstanding the dangers of the potholes, however, no past studies have developed cost-effective real-time roadway pothole detection and size estimation using built-in vehicle technologies in the reviewed literature.

This paper aims to develop simple, cost-effective real-time pothole detection technology to estimate pothole size and geolocation. The paper proposes the use of the deep learning (DL) approach to carry out the detection of potholes and the estimation of their sizes. The paper uses standard two-dimensional (2D) images to estimate a pothole's plan dimensions - length and width. These dimensions will help authorities estimate the quantity of pavement materials needed to repair them. The study aims to assist transportation agencies with real-time pothole detection as part of road condition monitoring (RCM), an essential part of road maintenance and rehabilitation (M&R). To that end, detecting and repairing potholes as they form will improve traffic safety and avoid economic losses to both governments and drivers.

Literature review

This section reviews the current state of the art for techniques used in Machine Learning (ML) and Deep Learning (DL) to carry out pothole detection and size estimation. Many past studies have used ML and DL technologies to detect and estimate the size of potholes. For example, past work by Motwani and Sharma (2020) used ML methods to carry out pothole detection and dimension estimation. The paper used K-nearest neighbor (KNN) in building the detection model. The study used both Manhattan and Euclidean distance measures to estimate the dimensions of the detected potholes. The detection model achieved 100 % accuracy in training and 89 % in testing using 150 and 50 images, respectively. Another study by Lin and Liu (2010) used support feature extraction and a non-linear vector machine (SVM) approach in pothole detection. The method used a partial differential equation (PDE) for image segmentation, and it could clearly distinguish potholes from other distresses from 30 images used for testing purposes. Also, another system was proposed to carry out pothole detection and estimate their dimensions based on the YOLOv4 algorithm for detection and a triangle similarity property to estimate pothole dimensions. The proposed detection model achieved 0.933 mean Average Precision at 50 % (mAP@0.5) and 0.741 Intersection over Union (IoU). Still, the size estimation module is prone to errors as it involves the manual setup of the camera, therefore providing room for improvements (Chitale et al., 2020).

A convolutional neural network (CNN) approach based on YOLOv3 was also employed to detect potholes (Hedeya et al., 2022). The study used 3D point clouds to classify the severity of the detected potholes based on region of interest (RoI) extraction. The detection model was trained using colored images from different sources and achieved excellent results. However, false negative (FN) detections were reported in poorly represented classes. The shortcoming of the proposed approach is due to the use of 3D point clouds in severity classification since these images are more expensive than 2D images and require more computational efforts.

Kharel et al. (2022) compared the real-time pothole detection performance of YOLOv5 Large (Y_l), YOLOv5 Medium (Y_m), and YOLOv5 Small (Y_s) using ResNet101 backbone and Faster R-CNN using ResNet50 (FPN), VGG16 and MobileNetV2 backbones. YOLOv5$_s$ outperformed the rest and proved the best for real-time pothole detection, with an average detection speed of 0.009 s per image. They also employed inverse perspective mapping (IPM) to estimate the plan dimensions of the potholes. This study provides an accurate and fast approach to detecting and estimating dimensions of potholes; however, it requires manual measurement of road width to determine the scale fed into the system, which may be prone to errors.

Furthermore, Zaimis et al. (2019) employed a DL algorithm to detect potholes from 2D colored images. The authors used semantic and instance segmentation approaches to carry out image processing. These techniques ensure that only part of the image containing a pothole will be processed, saving time and improving efficiency. The study then employed ultrasonic sensors to estimate the volumes of potholes. The sensors proved to be accurate; however, they are greatly affected by noises, affecting their accuracy. The sensors are also more expensive than standard 2D images which involve the application of DL (Majidifard et al., 2020).

DL approach was also used by Arjapure and Kalbande (2021), who employed the Mask Region-Based Convolutional Neural Network (Mask R-CNN) approach to detect potholes from 2D images. Region of Interest (RoI) was extracted and used to estimate the areas of potholes. The testing results showed that the approach attained an accuracy of 90 % between the estimated areas and the ground truth, with $a \pm 10$ % deviation. This approach is accurate but prone to errors as it involves manual scale estimation to obtain actual potholes' dimensions from images (Arjapure & Kalbande, 2021). According to recent research by Aware et al. (2023), the Mask R-CNN algorithm was developed to identify potholes and estimate their plan area. The proposed system reached an overall mean average precision (mAP@50) of 0.48b. Other studies including (Ahmed et al., 2022; Egaji, Evans et al., 2021; Maeda et al., 2018; Ruseruka et al., 2023a, 2023b) only aimed at developing modern and cheap methods for the detection of pavement distresses, with no dimension or severity considerations.

In summary, past studies in the literature have yet to attempt to estimate the size of the pothole and its geolocation based on built-in vehicle technologies. This paper proposes using built-in vehicle technologies to detect and determine pothole size and geolocation. Specifically, the study develops YOLOv5 to detect and measure the size of the potholes in real-time using the vehicle's built-in camera. The developed model will use a vehicle lane-keeping assistance (LKA) system to estimate the scale from the roadway automatically. The estimated scale is automatically incorporated into the detection model, providing the detected potholes with actual dimensions. The main contribution of this study is to provide a model that uses the vehicle built-in technologies already incorporated in some vehicle models to provide real-time detection and estimation of the size of the potholes and geolocation.

Methodology

This section presents a detailed description of the data used and methods used in real-time detection and estimation of the size of potholes using built-in vehicle technologies. Subsections that follow discuss data sources, preparation, model development, and predictions.

Data sources

This paper prepared the computer-vision detection model in four steps: dataset acquisition, data preparation, model training, and model testing. In data acquisition, the trained model is based on computer vision; therefore, it needs images for training and testing purposes. Various pothole images were collected from the internet for model training and testing. The sources of these images include the Dataset for Smartphone-based Road Damage Detection and Classification. This dataset contains 26,336 collected using smartphones mounted on car dashboards in Japan, India, and the Czech Republic (accessible through the link: https://doi.org/10.17632/5ty2wb6gvg) and another pavement distresses v12-v4 from Roboflow which contains 665 images (accessible through the link: https://public.roboflow.com/object-detection/pothole). A random sample of 1876 images was selected from the collected images for model preparation. The sample includes images with dry potholes, wet potholes (with water), daytime, and nighttime and had potholes with varying sizes.

The pothole dimension estimation module was prepared using the American Honda Motor Co., Inc. video dataset. The video dataset comprises about 104 h of real human driving scenarios and shows a record of roads across various streets in California State, in the United States. The videos were collected using an in-vehicle video camera. The dataset is identified as "*Toward Driving Scene Understanding: A Dataset for Learning Driver Behaviours and Causal Reasoning*" and it is available for non-commercial uses upon signing an agreement on the terms of use (Ramanishka, Chen, Misu & Saenko, 2018).

Data preparation

The first step in data preparation was to select a sample of images for model training. The preparation process involves image annotations, whereby the objects of interest are marked with bounding boxes in the images. This process is performed using an onlineakesense.ai tool (M.S., 2023). In this process, rectangular bounding boxes were drawn around the potholes in each image, and a pothole label was assigned. A total number of 20,715 instances were generated in this process. Fig. 1 illustrates the interface of the annotation tool. An augmentation technique was then employed to increase the data for training the model. This method is proper and used commonly when no sufficient dataset is available for model training and validation.

Data augmentation

Data augmentation is a technique to improve the training dataset to avoid overfitting. The method uses the available image dataset by increasing the number of cases from which the model will train. Commonly used methods include flipping, rotation, blur, Random zoom, vertical and horizontal shifts (translations), and brightness changes. Each technique produces several images from one image, increasing the number of instances. As a result, the model trains on more images, which helps to improve the accuracy, which is the function of the amount of data used (Indah et al., 2024).

Flipping

Flipping involves the reflection of an image over a straight line, either horizontally or vertically. Fig. 2 shows some of the images generated by applying this technique.

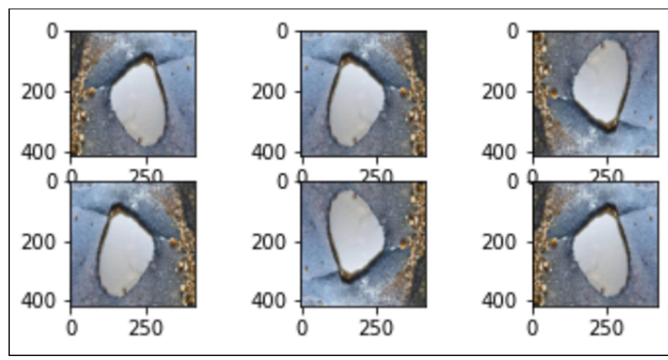

Fig. 2. Image flipping.

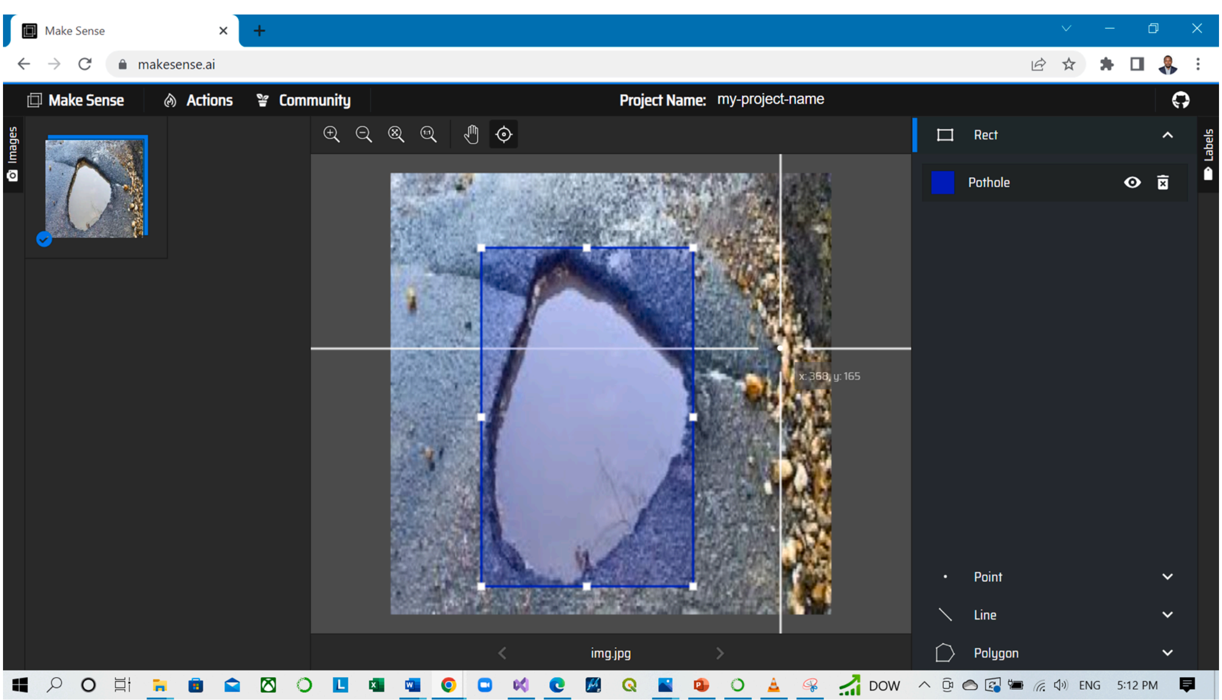

Fig. 1. Image annotation process.

Rotation

This technique increases the number of images by rotating the images randomly, thereby making several copies from one image. In this study, the rotation range value was set to 60. Fig. 3 shows the sample of images formed by random rotations of one image.

Blur

Image blurring makes an image less sharp by smoothing the color transition between the pixels. A blurred image cannot be seen clearly because its edges are no longer distinct. Fig. 4 shows some of the images formed from blurring.

Random zoom

Random zoom increases and decreases the object size randomly. This technique also provides more images from which the algorithm learns. Once the image has been decreased in size, the space left behind is filled by a 'nearest' option, which renders the space with nearby pixels from the image. This brings uniformity to the images. Fig. 5 shows some images formed from a random zooming technique.

Vertical and horizontal shifts

This technique applies vertical and horizontal transitions to the images to generate new copies. The spaces left behind in the images were filled by a 'nearest' technique to ensure that the spaces left behind resemble the nearby pixels to bring uniformity. Fig. 6 illustrates images created from vertical and horizontal shifts.

Brightness

The brightness technique involves the change of light intensity on images. When applied to the same image, various light intensities produce several copies of the same image, thereby providing more images from which the model learns during the training process. Fig. 7 shows sample images generated from a single image when the brighness range was between 0.2 and 1.0.

Pothole detection and size estimation

In this paper, the method used for detecting and estimating the size of the potholes is divided into two parts namely: 1) pothole detection and 2) pothole size estimation. Fig. 8 illustrates the architecture of the methodology employed in this study. These parts are integrated to aid the data collection process. The collected data can be accessed and stored in local storage devices through vehicle onboard diagnostics (OBD) systems. Through a partnership between the authorities and private vehicle owners, who are regular commuters of the given road sections, this data can be shared wirelessly with the authorities for further action.

Pothole detection model

Pothole detection involves the preparation of a model that detects

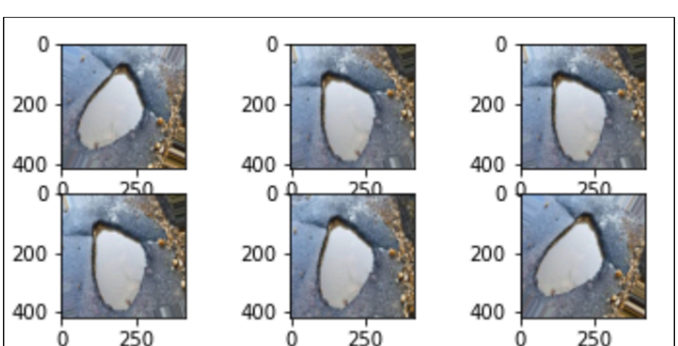

Fig. 3. Image rotations.

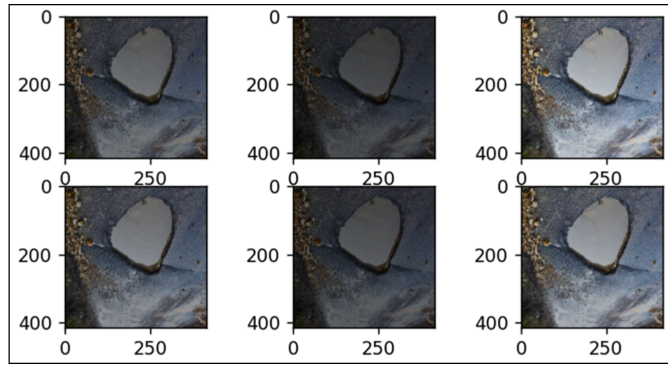

Fig. 4. Image blurring.

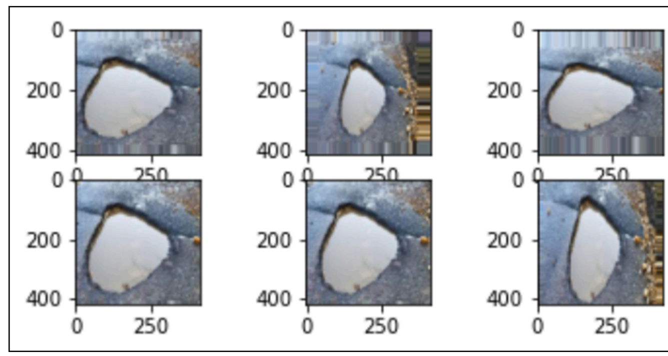

Fig. 5. Random zoom.

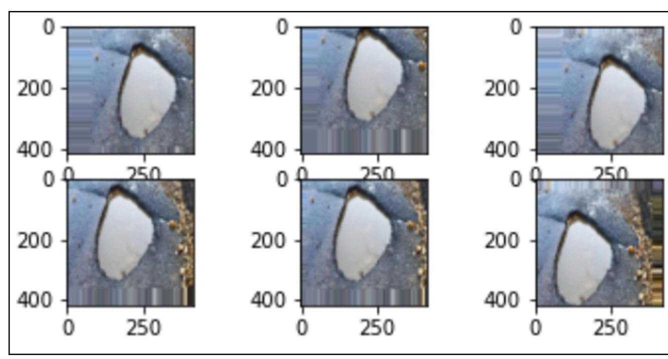

Fig. 6. Vertical and horizontal shifts.

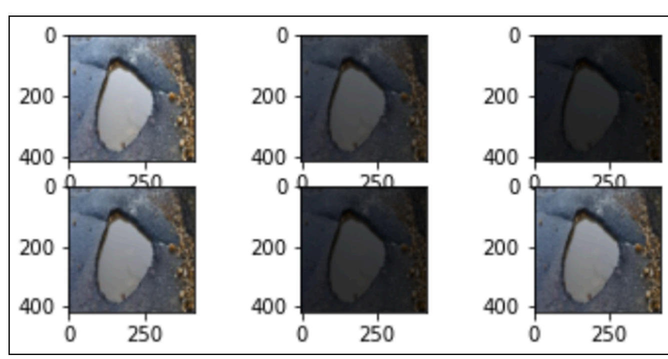

Fig. 7. Random brightness.

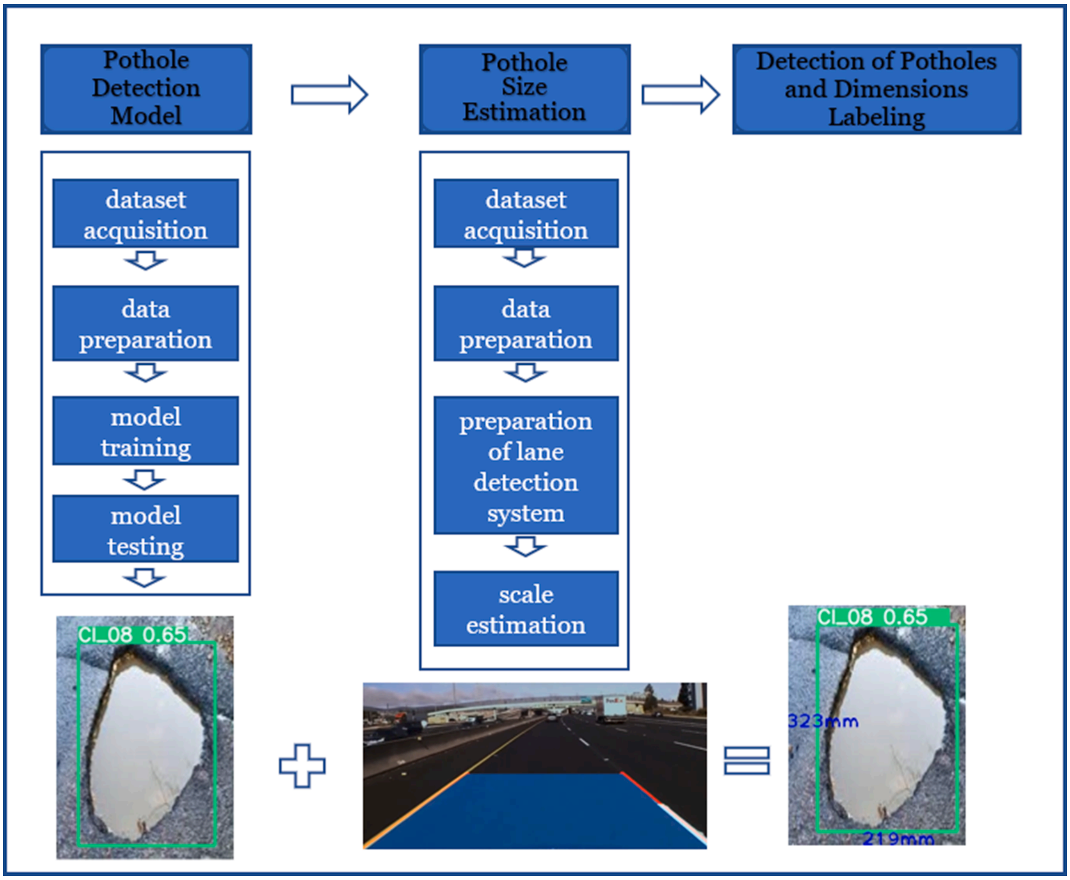

Fig. 8. Proposed system flowchart.

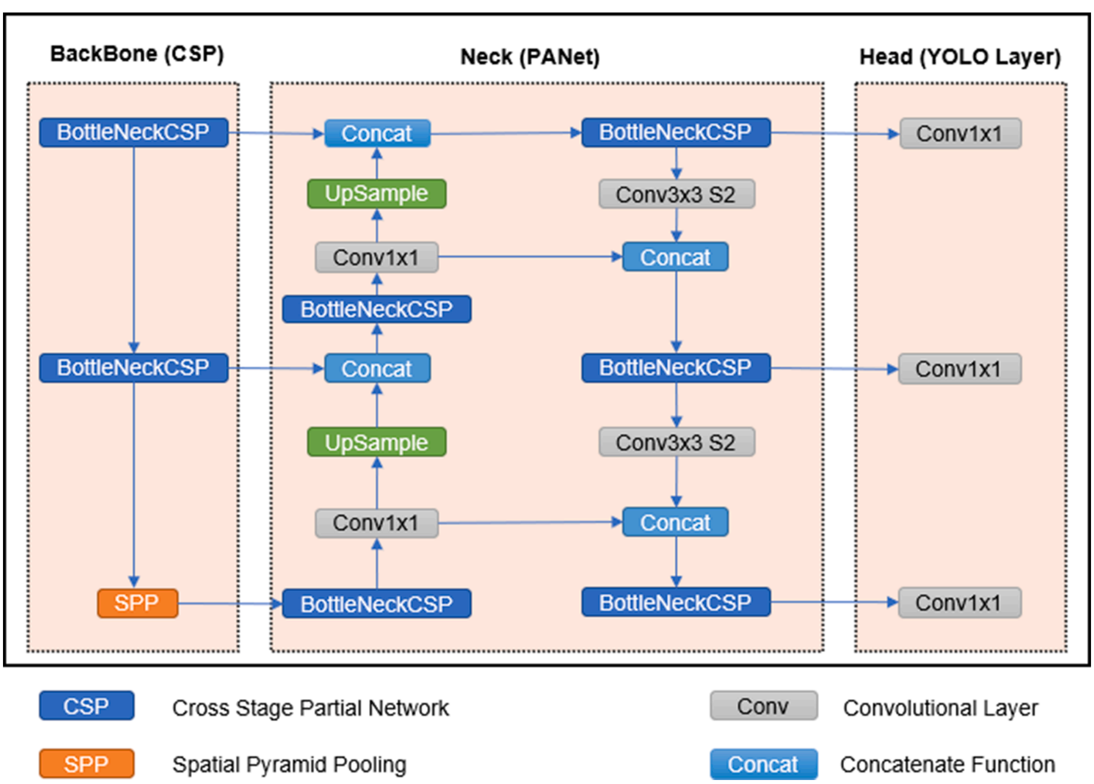

Fig. 9. YOLOv5 Architecture.

the potholes based on computer vision. In this study, You Only Look Once (YOLOv5) algorithm is employed to prepare the model. Fig. 9 shows the architecture of the YOLOv5 used in this paper. The YOLOv5 is a convolutional neural network (CNN) architecture designed for object detection, an essential task in computer vision. The network consists of 24 convolutional layers that extract features from the input image and then use these features to perform object detection using a set of head layers. The YOLOv5′s initial layers are designed to detect low-level image features such as edges and shapes. The filters become more complex and specialized as the layers progress to see more complex features. The spatial pyramid pooling layer of the architecture allows it to have variable-sized inputs and outputs, making it more adaptable for real-world applications.

The head layers are divided into the neck and the detection head. The neck comprises two convolutional layers that refine the features generated by the backbone layers. The detection head uses the neck's output from several convolutional layers to predict the bounding box coordinates, objectness scores, and class probabilities. The final classification head takes the detection head's output and predicts the class. This output allows the network to detect multiple objects in an image and classify them into different categories. The final model predictions are stored in the output layer, which includes the bounding box coordinates, objectness scores, and class probabilities.

Model training. The computer vision model in this paper was trained on the Google Colaboratory (Google Colab) environment. The online model training platform provides a free domain with up to 32GB of memory for training. Before training, the image dataset was split into 80 % to 20 % ratios for training and validation/testing, respectively. The YouOnly Look Once Version V (YOLOv5) algorithm trained the model. A total of 165 images with no potholes were included in the training dataset. These images are called background images and are included in the dataset to reduce the effects of false positives (FPs) and false negatives (FNs), thereby improving the model performance. The model was trained for 2.05 h on 140 epochs, which was determined to prevent overfitting, a phenomenon observed when validation losses start to rise. The training time is a function of the selected number of epochs, all things kept constant. A default stochastic gradient descent (SGD) of 0.01 was used, and anchor sizes were set to dynamic. The dynamic setting allows anchor boxes (used to detect objects of various sizes and shapes) to be adjusted automatically based on the dataset's characteristics during training. Also, a batch size of 20 was used, a value that has a good trade-off between memory usage and training speed (Chengula et al., 2023b; Isa et al., 2022; Trupti et al., 2021).

Model performance evaluation. The performance of the proposed model is assessed based on the evaluation of performance metrics, namely precision, recall, F1-score, and mean average precision (mAP). Precision measures how the model predicts the distresses correctly, whereas recall measures how the model detects all the distresses in the images. The following relationships, in Eqs. (1) and 2 give precision and recall values, respectively, which are the functions of true positive (TP), false positives (FP), and false negative(FN).

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

False Positives (FPs) and False Negative (FNs) are regarded as type I and type II errors, respectively. The FPs show how the model mispredicts pavement distress, whereas the FNs show how it fails to recognize them in images. On the other hand, the F1-score combines precision (P) and recall (R). It is obtained as a harmonic mean of the two and accounts for the differences in class sizes, i.e., class-imbalanced data distribution. Its values range from 0 to 100 %, and higher values of the F1-score denote a better model performance. The F1-score is obtained as shown in Eq. (3).

$$F1 - score = \frac{1}{\frac{1}{2}\left(\frac{1}{P} + \frac{1}{R}\right)} = \frac{2 * P * R}{P + R} \tag{3}$$

The mAP is the mean (average) of average precisions of all individual classes in the model. It is calculated from Eq. (4) below, where AP_k stands for Average Precision of class *k*, and *n* stands for the total number of classes.

$$mAP = \frac{1}{n}\sum_{k=1}^{k=n} AP_k \tag{4}$$

Model training results and discussion. Table 1 summarizes the obtained model training results. The values attained in this study are higher than the values from previous studies. For instance, the past that aimed at detecting pavement potholes using simple image processing techniques such as a Canny filter and contour detection attained 81.8 % and 74.4 % values in precision and recall, respectively (Nienaber, Booysen & Kroon, 2015). The value of mAP obtained is higher than 93 % which was achieved in a newly published study (Chitale et al., 2020) and the value of the F1-score is higher than 73.14 % which was attained in a study that proposed a crack and pothole detection system based on deep neural network architecture (Anand, Gupta, Darbari & Kohli, 2018).

Pothole size estimation model

This section developed the means to estimate the actual dimensions of potholes. It is inspired by the Lane Keeping Assistance (LKA) system of modern vehicle models. The system is trained to 'see' lanes on which the vehicle passes, using the Hough lines in the OpenCV library in Python and measuring the distance between the two lanes, as shown in a pseudocode in Algorithm 1. The scale generation used in this study using the OpenCV library is illustrated in Fig. 10.

The distance between the lines is measured perpendicularly to both left and right lane lines using the Euclidean formula shown in Eq. (5).

$$d(p, q) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2} \tag{5}$$

Where d(p, q) is the distance between two points p and q located along the detected lane lines, with coordinates (p_1, p_2) and (q_1, q_2), respectively. This formula gives the output in pixels; therefore, it is necessary to convert them to get convenient dimension units to be used to estimate pothole sizes. Eq. (6) shows a mathematical expression of how the conversion was done by considering the actual dimensions of lanes (*L*) in millimeters, which should be determined prior to the data collection process. The perpendicular distance (*d*) between the lane lines gives the lane width in pixels, which is then used to obtain the dimensions of pixels.

$$pixel\ size\ (scale) = \frac{L}{d(p,q)} \tag{6}$$

The output of Eq. (6) provides the scale fed directly into the pothole detection model to get the actual dimensions of potholes in millimeters. For example, for a road segment with 3600 mm wide lanes and a length of 10,975.61 pixels determined by Eq. (5), a 0.328 mm/pixel scale was obtained. Both values calculated in Eqs. (5) and 6 were then fed into the pothole detection model to provide the actual lengths and widths of the

Table 1
Model training results.

S/N	Metric	Value
1.	Precision	93.0
2.	Recall	91.6
3.	F1-score	87.0
4.	mAP	96.3

Algorithm 1
Pseudocode for Hough Lines Transform.

```
# Step 1: Convert the image to grayscale
gray_image = convert_to_grayscale(original_image)
# Step 2: Apply edge detection (to detect lane marking)
edges = apply_canny_edge_detector(gray_image)
# Step 3: Initialize the Hough accumulator matrix
accumulator_matrix = initialize_accumulator(edges)
# Step 4: Define the parameter space (rho, theta)
rho_resolution = 1 # Resolution for rho parameter
theta_resolution = 1 # Resolution for theta parameter
# Step 5: Loop through each edge pixel in the image
for edge_pixel in edges:
# Step 6: Loop through each possible theta value
for theta in range(0, 180, theta_resolution):
# Step 7: Calculate the corresponding rho value
rho = edge_pixel.x * cos(theta) + edge_pixel.y * sin(theta)
# Step 8: Increment the accumulator matrix at the (rho, theta) position
accumulator_matrix[rho, theta] += 1
# Step 9: Define a threshold to identify significant peaks in the accumulator matrix
threshold = 50
# Step 10: Extract lines from significant peaks in the accumulator matrix
lines = extract_lines(accumulator_matrix, threshold)
# Step 11: Draw the detected lines on the original image
result_image = draw_lines(original_image, lines)
```

Fig. 10. Scale generation using OpenCV library.

potholes in millimeters.

Camera transformations. Images taken by built-in vehicle cameras are inclined at an angle to the horizon. Top views of the images taken are necessary to get the actual dimensions of potholes. Therefore, we used the inverse perspective mapping (IPM) technique to address this problem (Kharel & Ahmed, 2022). This technique projects the images to get views perpendicular to the surface (bird's eye view). This study used the OpenCV library in Python to carry out IPM. The IPM can be achieved using two methods, namely, i) using intrinsic camera parameters and ii) using four-point perspective transforms (Gerum et al., 2019). The intrinsic camera parameters technique performs well with the moving camera, whereas the four-point perspective transforms perform well with static images. This study used the intrinsic camera methods because we analyzed the moving images. Focal length, sensor size, image size, tilt degree, roll degree, and elevation parameters are necessary for perspective transformation. Fig. 11 illustrates an image of a pothole before and after applying the IPM transformation technique.

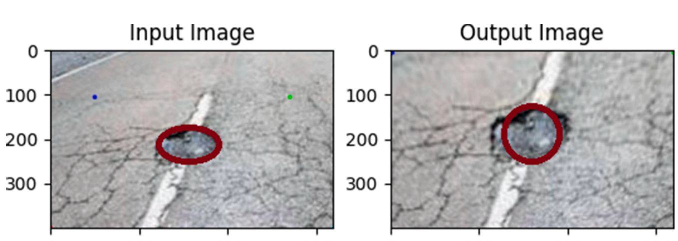

Fig. 11. Illustration of the before and after application of the IPM technique.

Inference on testing images

The trained pothole detection model was run on testing images to measure its performance in predicting the pothole dimensions. The testing set included images the model had not seen before. Fig. 12 shows sample outputs indicating pothole markings, measurements, and corresponding detection confidences.

Model validation

The performance of the developed model was tested by observing the dimensions of the detected loop detectors. The estimated sizes were compared to the actual measurement of the loops, determined by the California Department of Transportation, where the video dataset used in this study was recorded (USDoT, 2016). Fig. 13 shows the dimensions of detection loops as estimated by the model.

Table 2 summarizes the actual and predicted measurements of the detection loops. Before applying the scale, the dimensions of the loop were 5454.06 by 5452.36 pixels. Since the detection loops were square (1828.8 mm x 1828.8 mm), the estimated width was used to calculate the longitudinal dimension of the pixels. Then, this value provided the scale used to estimate the plan widths of potholes fed into the detection system to give the estimated width of potholes in millimeters. Eqs. (7) and 8 indicate the computation of average width and longitudinal scales used in the model.

$$Average\ width = \frac{width_1 + width_2}{2} = \frac{1769.00 + 1775.03}{2} = 1772.02mm \tag{7}$$

$$Longitudinal\ Scale = \frac{1772.02}{5452.36} = 0.325\ mm \Big/ pixel \tag{8}$$

The detection was extended to collect data for statistical analysis. Twenty predictions were recorded as seen in Table 3. The predicted values obtained had a mean of 1828.032, a standard deviation of 26.20, and were significant at a 5 % level (p-value = 0.037).

Conclusion

This study proposed using built-in vehicle technologies to detect and estimate pothole measurements. The model estimation process involved

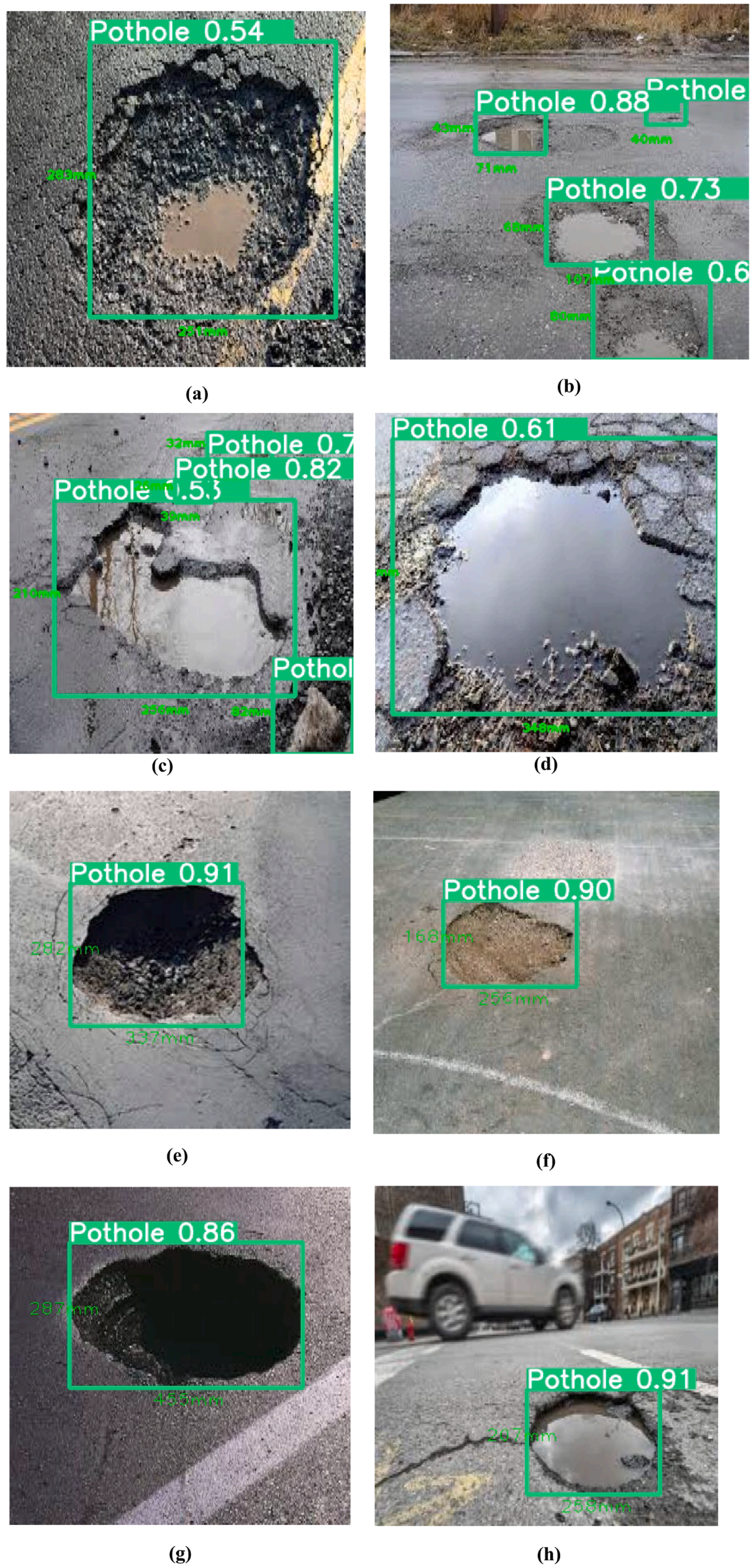

Fig. 12. Sample model output measurements.

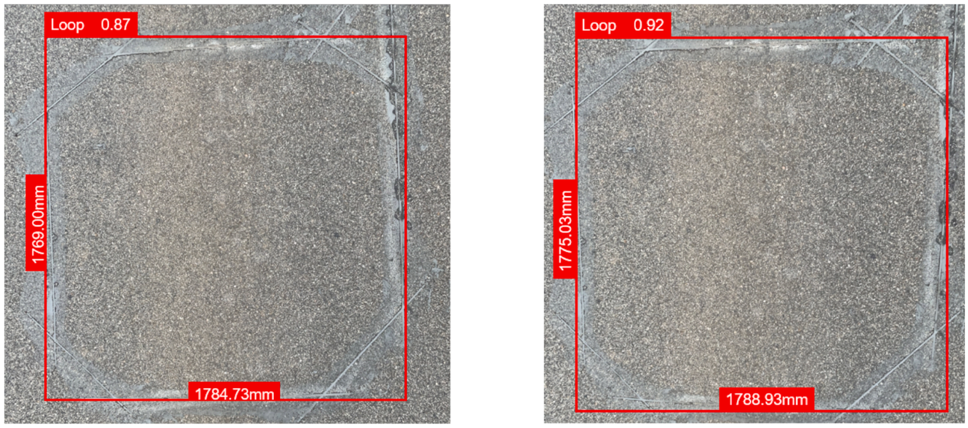

Fig. 13. Dimensions of detection loops as estimated by the model.

Table 2
Model Testing Results.

S/N	Actual size (mm)		Estimated size (mm)	
	Length	Width	Length	Width
1.	1828.8	1828.8	1784.73	1769.00
2.	1828.8	1828.8	1788.93	1775.03

Table 3
Sample Prediction Results.

S/N	Length (mm)	Width (mm)	S/N	Length (mm)	Width (mm)
1	1826.94	1827.07	6	1828.72	1829.31
2	1831.01	1826.48	7	1784.73	1788.93
3	1827.32	1832.21	8	1829.3	1826.85
4	1830.89	1826.06	9	1832.28	1830.33
5	1827.56	1828.12	10	1829.43	1826.42

training 1876 normal 2D colored pothole images and predicting estimating measurements. The detection model was prepared using the YOLOv5 algorithm and trained on Google Colab for 2.05 h. The pothole training images were manually annotated because there is no trained pavement distress detection for use on the YOLO models. The trained model attained 93.0 %, 91.6 %, 96.3 %, and 87.0 % in precision, recall, mean average precision, and F1-measure metrics, respectively. These results agree with the recently published works in the state of the art, including (Anand et al., 2018; Chitale et al., 2020).

Furthermore, the paper used Hough lines in the OpenCV Python library to detect pavement line markings and compute pothole measurements. This study used many modern vehicles' standard lane-keeping assistance (LKA) systems of many modern vehicles to establish pothole measurement scales. This paper created width and longitudinal scale from known measurements of the two loop detectors in the same dataset. The comparison results between the actual and estimated sizes indicated that the proposed model can estimate the size of potholes with a 2.30 % error. This result suggests that built-in vehicle technologies, like the camera and LKA system, can provide cheap ways to estimate pothole dimensions. This study can help road authorities collect pothole data cheaply in real-time by utilizing built-in vehicle technologies available in many standard vehicles.

Study limitations

This study proposes a method that uses built-in vehicle technologies to determine pothole sizes. This method develops a scale using the Hough line technique from the Python OpenCV library. This technique detects feature edges such as pavement lane lines and measures the perpendicular distance between them to estimate the scale. Thus, this technique could be used only on roads whose lanes are marked and visible.

Recommendations

This study used a scale computed from pavement lane markings to generate the width of a pothole. There is a need to research another scale-generation approach for applying to roads without marked lanes. One option is to use standard objects and devices on roadways whose dimensions are known for scaling pothole dimensions.

CRediT authorship contribution statement

Cuthbert Ruseruka: Conceptualization, Data curation, Methodology, Writing – original draft. **Judith Mwakalonge:** Conceptualization, Methodology, Project administration, Supervision. **Gurcan Comert:** Conceptualization, Data curation, Methodology. **Saidi Siuhi:** Writing – review & editing. **Frank Ngeni:** Conceptualization, Data curation. **Quincy Anderson:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This study used image and video datasets collected from the internet and is freely available. The video dataset was collected from Honda Motor Company and is made available upon request and signing the agreement of their use (Ramanishka et al., 2018).

Acknowledgement

U.S. Department of Education partly supports this study through the HBCU Master's Program Grant, by Grant No. P120A210048, the U.S. Department of Transportation's University Transportation Centers Program grant administered by the Transportation Program at South Carolina State University (SCSU), Tier I University Transportation Center for Connected Multimodal Mobility, and NSF Grant Nos. 1719501, 1954532, and 2131080.

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